

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 3: Filtering & Spatial Domain Operations

PART 1: MATCH THE FILTER WITH ITS PURPOSE

- **Mean Filter** → C. Smooths image
- **Median Filter** → A. Removes salt & pepper noise
- **Laplacian** → B. Image sharpening
- **Gaussian Filter** → D. Noise reduction with weighting

PART 2: EDGE PRESERVATION ANALYSIS (MEAN VS. MEDIAN)

Which filter preserves edges better?

The **Median Filter** preserves edges significantly better than the Mean Filter.

Why?

- **Non-linear Operations:** The median filter is a non-linear order-statistic filter. It selects the median intensity value from the local neighborhood rather than calculating an arithmetic average.
- **Prevention of Intensity Blurring:** A mean filter averages pixel values across an edge boundary, creating intermediate pixel values that blur sharp transitions between regions.
- **Outlier Elimination:** Because extreme impulse pixel values (such as salt & pepper noise) fall at the high and low ends of the sorted array, the median operation completely discards them without letting them shift or blur the true boundary values across edges.